

Sliding Window Flow Control

Flow control in computer networks ensures that the sender does not overwhelm the receiver. Sliding Window is an efficient technique that allows multiple frames to be sent at once before receiving an acknowledgment (ACK).

Concept of Window

- A window is the range of frames that can be sent or received at a given time.
- Sender Window → Frames that the sender can send.
- Receiver Window → Frames that the receiver can accept.

Working of Sliding Window

1. At the beginning, the sender can send a number of frames equal to the window size.
2. When an acknowledgment (ACK) is received, the window slides forward.
3. The sliding allows the sender to transmit the next new frame.

Example (Window Size = 4, Frames 0–7)

1. Initial Window:

[0][1][2][3] → Can send now

[4][5][6][7] → Must wait

2. If ACK(0) arrives → Window slides by 1:

[1][2][3][4] → Frame 4 is new to send

3. If ACK(1) arrives → Window slides by 1:

[2][3][4][5] → Frame 5 is new to send

4. If cumulative ACK(4) arrives → Window slides by 4:

[4][5][6][7] → All are new to send

Types of Sliding Window Protocols

- Go-Back-N ARQ → If error occurs, retransmit from the erroneous frame onward.
- Selective Repeat ARQ → Only the erroneous/lost frame is retransmitted; others are stored.

Key Points

- At the beginning, sender sends all frames in the window.
- Afterwards, for each ACK, usually only one new frame is sent (unless cumulative ACKs slide multiple steps).
- Cumulative ACKs can make the window slide many steps at once.
- Sliding Window ensures efficient use of bandwidth compared to Stop-and-Wait.